

**Problem 1.**

**Airport data:**

- What is the  $\alpha$  if the distribution is power law?
  - What is the  $\lambda$  if the distribution is exponential?
  - What is  $[a, b]$  if the distribution is uniform?
  - What is  $\mu$  and  $\sigma$  if the distribution is normal?
- (These can be CDF, PDF, or QQplot)
- Screen shot of the visualization of distribution of data
  - Screen shot of the visualization of distribution of data if it was power law
  - Screen shot of the visualization of distribution of data if it was exponential
  - Screen shot of the visualization of distribution of data if it was uniform
  - Screen shot of the visualization of distribution of data if it was normal
  - What is the distribution of data based on your observation? Discuss

### Movie Rating data:

- What is the  $\alpha$  if the distribution is power law?
  - What is the  $\lambda$  if the distribution is exponential?
  - What is  $[a, b]$  if the distribution is uniform?
  - What is  $\mu$  and  $\sigma$  if the distribution is normal?
- (These can be CDF, PDF, or QQplot)
- Screen shot of the visualization of distribution of data
  - Screen shot of the visualization of distribution of data if it was power law
  - Screen shot of the visualization of distribution of data if it was exponential
  - Screen shot of the visualization of distribution of data if it was uniform
  - Screen shot of the visualization of distribution of data if it was normal
  - What is the distribution of data based on your observation? Discuss

**Problem 2.**

- Screenshot of distribution/histogram of the age
- Screenshot of distribution/histogram of the first digit of age
- Screenshot of distribution/histogram of the last digit of age
- Discussion (what are those distribution?) Why do you think it is like this?